WORKED SOLUTION

STEP III 1991, Question 11

Mechanics • Rigid Body Equilibrium

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Read the question before you go on. It is in the STEP Support Programme's question database:

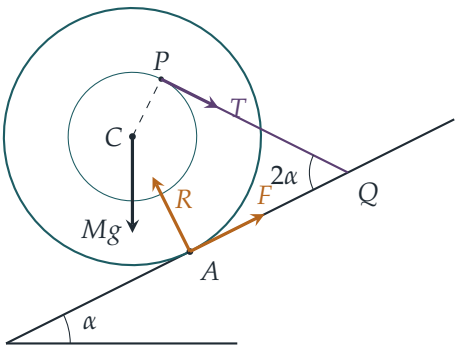
step.maths.org/questions/91-s3-q11

The key idea

The cylinder is a rigid body in equilibrium under four forces: its weight, the normal reaction, friction at the point of contact, and the tension in the string. Two of these, the weight and the normal reaction, pass through the centre of the cylinder, so taking moments about the centre links the friction directly to the tension. Resolving along and perpendicular to the plane then gives the friction and the normal reaction, and the inequality $F \leq \mu R$ does the rest. In the second part the cylinder doesn't know what is holding the string, so the tension is unchanged and we only need to look at the particle.

Solution

We model the cylinder as a uniform rigid body whose centre of mass is its centre C, the string as light and inextensible, and the second body as a particle. Let A be the point of contact with the plane, and let the string's tension be T. Since the string is wound round the groove, the free part PQ is tangent to the groove at P, so its line of action is at perpendicular distance a from C. From the diagram, Q is further up the plane than A, and the string pulls the cylinder from P towards Q: up the plane and in towards it, at angle 2α to the plane.



The dashed radius CP has length a and is perpendicular to the string. We take F as the friction on the cylinder, positive up the plane.

The cylinder: showing $\mu \geq \frac{1}{3}\tan\alpha$

Moments about C. The weight acts at C and the normal reaction acts along AC, so neither has a moment about C. Friction acts at distance $2a$ from C and the tension at perpendicular distance a, turning the cylinder in opposite senses, so

$$F \cdot 2a = T \cdot a, \qquad \text{that is,} \qquad T = 2F.$$

Ask yourself Why do the friction and the tension turn the cylinder in opposite senses?

They are the only two forces with a moment about C, so their moments must cancel. On its own, the string's pull at P would turn the cylinder so that it rolls up the plane. Friction at A pointing up the plane turns it the other way. If we had guessed the direction of friction wrongly, the equations below would simply give $F < 0$, so the sign isn't an extra assumption.

Resolving parallel to the plane (up the plane positive):

$$F + T\cos 2\alpha = Mg\sin\alpha \quad \Longrightarrow \quad F(1 + 2\cos 2\alpha) = Mg\sin\alpha.$$

Resolving perpendicular to the plane: the tension has a component $T\sin 2\alpha$ into the plane, so

$$R = Mg\cos\alpha + T\sin 2\alpha = Mg\cos\alpha + 2F\sin 2\alpha.$$

Substituting $F = Mg\sin\alpha/(1 + 2\cos 2\alpha)$,

$$\begin{aligned} R(1 + 2\cos 2\alpha) &= Mg(\cos\alpha + 2\cos 2\alpha\cos\alpha + 2\sin 2\alpha\sin\alpha) \\ &= Mg(\cos\alpha + 2\cos(2\alpha - \alpha)) = 3Mg\cos\alpha. \end{aligned}$$

So

$$F = \frac{Mg\sin\alpha}{1 + 2\cos 2\alpha}, \qquad R = \frac{3Mg\cos\alpha}{1 + 2\cos 2\alpha}.$$

The cylinder stays in contact, so $R > 0$. Since α is acute, $\cos\alpha > 0$, and so $1 + 2\cos 2\alpha > 0$, which means $F > 0$ as well. Dividing,

$$\frac{F}{R} = \frac{\sin\alpha}{3\cos\alpha} = \tfrac{1}{3}\tan\alpha.$$

For equilibrium friction must satisfy $F \leq \mu R$, so $\mu \geq F/R = \frac{1}{3}\tan\alpha$, as required.

The particle: the least coefficient of friction

The string stays where it was, so the cylinder is in exactly the same position and its three equations are unchanged. In particular the tension is still

$$T = 2F = \frac{2Mg\sin\alpha}{1 + 2\cos 2\alpha}.$$

With $\tan\alpha = \frac{1}{2}$ we have $\sin\alpha = \frac{1}{\sqrt{5}}$ and $\cos\alpha = \frac{2}{\sqrt{5}}$, so

$$\cos 2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \frac{3}{5}, \qquad \sin 2\alpha = \frac{2\tan\alpha}{1+\tan^2\alpha} = \frac{4}{5},$$

and $1 + 2\cos 2\alpha = \frac{11}{5}$. Hence

$$T = \frac{2Mg}{\sqrt{5}} \cdot \frac{5}{11} = \frac{2\sqrt{5}}{11}\,Mg.$$

(The cylinder also needs $\mu \geq \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.)

Now look at the particle at Q. The string pulls it towards P, that is, *down* the plane with component $T\cos 2\alpha$ and *away from* the plane with component $T\sin 2\alpha$. Its weight $3Mg$ also has a component down the plane, so friction F' on the particle must act up the plane. Let the normal reaction on it be R'.

Parallel to the plane:

$$F' = 3Mg\sin\alpha + T\cos 2\alpha = \frac{3Mg}{\sqrt{5}} + \frac{3}{5}\cdot\frac{2\sqrt{5}}{11}Mg = \frac{33\sqrt{5}}{55}Mg + \frac{6\sqrt{5}}{55}Mg = \frac{39\sqrt{5}}{55}Mg.$$

Perpendicular to the plane:

$$R' = 3Mg\cos\alpha - T\sin 2\alpha = \frac{6Mg}{\sqrt{5}} - \frac{4}{5}\cdot\frac{2\sqrt{5}}{11}Mg = \frac{66\sqrt{5}}{55}Mg - \frac{8\sqrt{5}}{55}Mg = \frac{58\sqrt{5}}{55}Mg.$$

This is positive, so the particle does stay on the plane. Equilibrium needs $F' \leq \mu' R'$, where μ' is the coefficient of friction between the particle and the plane, so

$$\mu' \geq \frac{F'}{R'} = \frac{39}{58}.$$

The least value of the coefficient of friction is $\dfrac{39}{58}$.

Checks

- Taking moments about A instead gives $T\,a(1 + 2\cos 2\alpha) = 2a\,Mg\sin\alpha$, the same tension. (The perpendicular distance from A to the string is $a + 2a\cos 2\alpha$.)
- As $\alpha \to 0$ the plane is horizontal, the tension and friction vanish, and the bound $\frac{1}{3}\tan\alpha \to 0$, as it should.
- A numerical check builds the geometry from coordinates for 10 000 random acute α below 60°, solves the equilibrium equations, and confirms $F/R = \frac{1}{3}\tan\alpha$. For $\tan\alpha = \frac{1}{2}$ it gives $T \approx 0.4066\,Mg$, $F' \approx 1.5856\,Mg$, $R' \approx 2.3580\,Mg$, and $F'/R' \approx 0.67241 = 39/58$.

Common slips

- Forgetting that the tension pushes the cylinder *into* the plane, which increases R. Leaving out $T\sin 2\alpha$ gives the wrong ratio.
- Getting the direction of the string's pull on the particle wrong. It pulls the particle towards P, so it *reduces* the normal reaction on the particle and adds to the down-slope force.
- Recomputing the tension from scratch for the second part. The cylinder's equilibrium hasn't changed, so neither has T.

Beyond STEP

Statics of rigid bodies comes down to two vector equations: the total force is zero and the total moment about *any* point is zero. Choosing that point well, here the centre, through which two unknown forces pass, is the main skill. At university this becomes the study of systems of forces and their equivalent "wrenches". Friction problems also lead on to questions about *how* equilibrium is lost, whether by slipping, rolling or toppling. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and rotation carefully.

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